

Distilling an LLM Diagram-Curation Pipeline into Local Classifiers

A practical study on one MacBook: data creation, a model ladder,
measurements, and a production decision

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Abstract

A frozen SigLIP2 image encoder with a logistic-regression head classifies figure crops as diagram / not-diagram at **86.0% against hand-verified truth** (95% CI [81.5, 90.5]), beating the vision-LLM pipeline that *generated its training labels* by **10.5 points** (75.5%; McNemar $p = 0.0005$), at **~52 ms/crop on a laptop GPU** versus 4–17 s per LLM call — offline, deterministic, at zero marginal cost. Zero-shot SigLIP2 (no training) also beats the LLM (83.5%, statistically tied with the trained probe). The complex options lost: an end-to-end fine-tune (71.5%) and MLP heads (78.0%) overfit the teacher’s ~25% label noise, and an off-the-shelf document layout detector for the page gate scored 61.7% while running 100× slower. We release the pipeline, the 2,000-diagram dataset, and all code. The central methodological lesson: under LLM distillation with noisy labels, validate against a hand-verified gold set or you will ship the wrong model — silver validation ranked the worst-generalizing model first.

1 Problem and contribution

We curate a dataset of *proper technical diagrams* (block diagrams, schematics, flowcharts, architectures — not data plots, photos, or fragments) from arXiv papers in six engineering fields, each image paired with a detailed label. Our production pipeline (v2) routes every page and every extracted crop through paid vision-LLM calls for two *classification* decisions — a page gate (“does this page contain a diagram?”) and a crop gate (“is this a proper diagram?”) — plus a genuinely *generative* labeling step. This paper asks whether the two classification gates can be replaced by local models on a laptop at equal-or-better accuracy and far better speed. The answer is yes, decisively, and the winning model is almost the simplest thing one can build.

2 How the data was created

Six arXiv field queries yielded 1,183 downloaded papers. A cheap screening model (**seed-1.6-flash**) proposes candidate diagram pages; a stronger model (**xiaomi/mimo-v2.5**) confirms; Mistral OCR extracts figure crops from confirmed pages; and **mimo-v2.5** classifies-and-labels each crop with strict rejection of plots, photos, fragments, and blanks. Free local pre-filters skip zero-graphics pages and tiny/blank crops. The run produced **2,000 accepted diagrams**, 3,868 rejected crops (with reasons), and 12,266 LLM-judged pages, for \$20.36 total, fully resumable (it survived credit exhaustion and two hard shutdowns without loss). Figure 1 shows the funnel.

Because every decision was journaled to SQLite, the classifier training sets are a byproduct: crops with 4-class silver labels (diagram / data_plot / photo / fragment), pages with binary silver labels, and a paper-level 70/15/15 split (no paper crosses splits).

Data provenance: how 1,183 arXiv papers became 2,000 labeled diagrams (pipeline cost \$20.36, fully resumable)

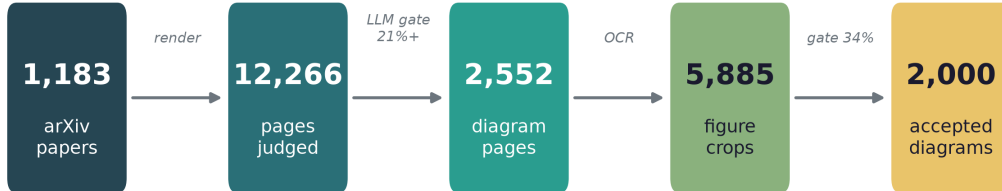


Figure 1: Data provenance. The pipeline is the “teacher”; its decisions become the students’ training labels.

3 Gold labels: the measurement instrument

Silver labels are LLM opinions. All headline numbers use **gold** labels: 200 test-split crops and 120 test-split pages, labeled *blind* against a versioned rubric by two independent annotators (Claude Sonnet agents with no access to model outputs), disagreements adjudicated by the lead. Inter-rater reliability was $\kappa = 0.925$ (crops, 4-class) and 0.940 (binary) — a sound instrument. Critically, **the LLM teacher itself scores only 75.5% on gold crops and 80.0% on gold pages**: it systematically over-accepts annotated plots and photos (precision 0.675 / 0.600). A student can therefore beat its teacher, and “agreement with the LLM” is not the same as accuracy.

4 The model ladder and results

We ran approaches in ascending complexity, each justified only if it beat the previous. Table 1 gives the gold-test headline.

Table 1: Crop gate, binary accept/reject vs. hand-verified gold ($n = 200$).

Model	Gold acc	95% CI	F1	Prec.	Rec.
SigLIP2 + logreg	0.860	[0.815, 0.905]	0.854	0.812	0.901
Zero-shot SigLIP2	0.835	[0.785, 0.885]	0.837	0.759	0.934
SigLIP2 + MLP	0.780	[0.725, 0.835]	0.796	0.688	0.945
DINOv2 + MLP	0.765	[0.710, 0.825]	0.773	0.690	0.879
LLM teacher (mimo-v2.5)	0.755	[0.700, 0.815]	0.768	0.675	0.890
MobileCLIP-S0 + MLP	0.750	[0.695, 0.810]	0.766	0.667	0.901
DINOv3 + MLP	0.735	[0.675, 0.795]	0.762	0.644	0.934
EfficientNet-B0 fine-tune	0.715	[0.650, 0.775]	0.695	0.677	0.714
Heuristics (GBT)	0.625	[0.555, 0.695]	0.678	0.556	0.868

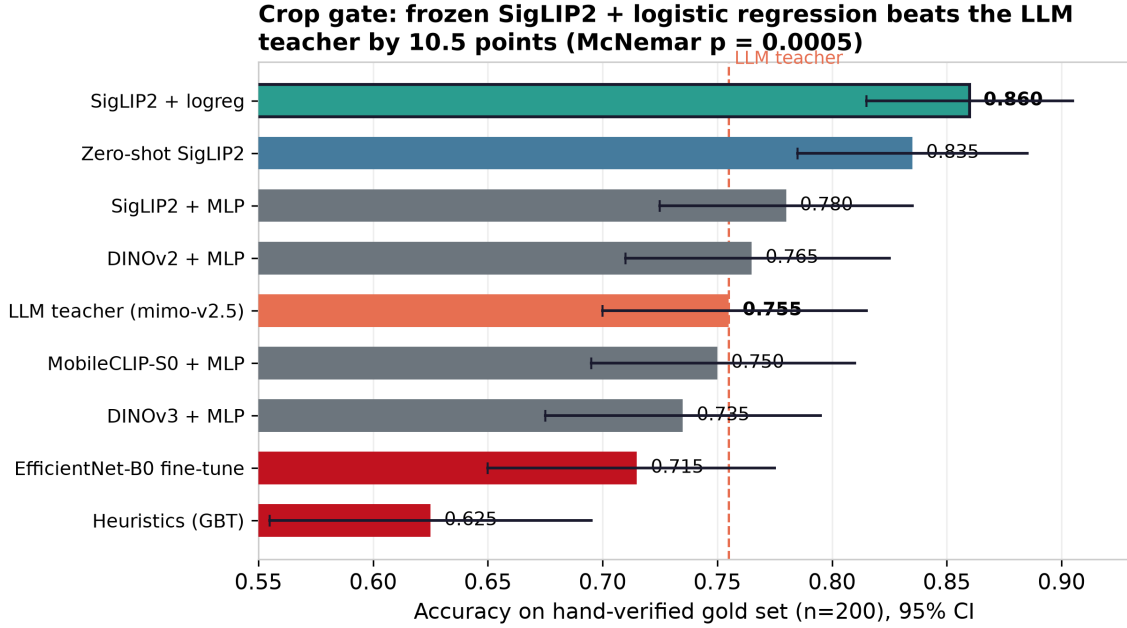


Figure 2: The winner beats the LLM teacher by 10.5 points; the gap is significant (McNemar $p = 0.0005$).

Three findings. (1) A frozen SigLIP2 backbone with a logistic-regression head beats the LLM teacher by 10.5 points, with McNemar $p = 0.0005$ (27 cases it fixes vs. 6 it breaks). (2) *Simplicity wins under label noise*: the models that topped silver-validation — MLP heads and the fine-tune — *fell* on gold because they had the capacity to memorize the teacher’s $\sim 25\%$ wrong labels; the un-overfittable models generalized (Figure 3). (3) Zero-shot, with no training at all, also beats the teacher and is statistically tied with the trained probe (McNemar $p = 0.38$).

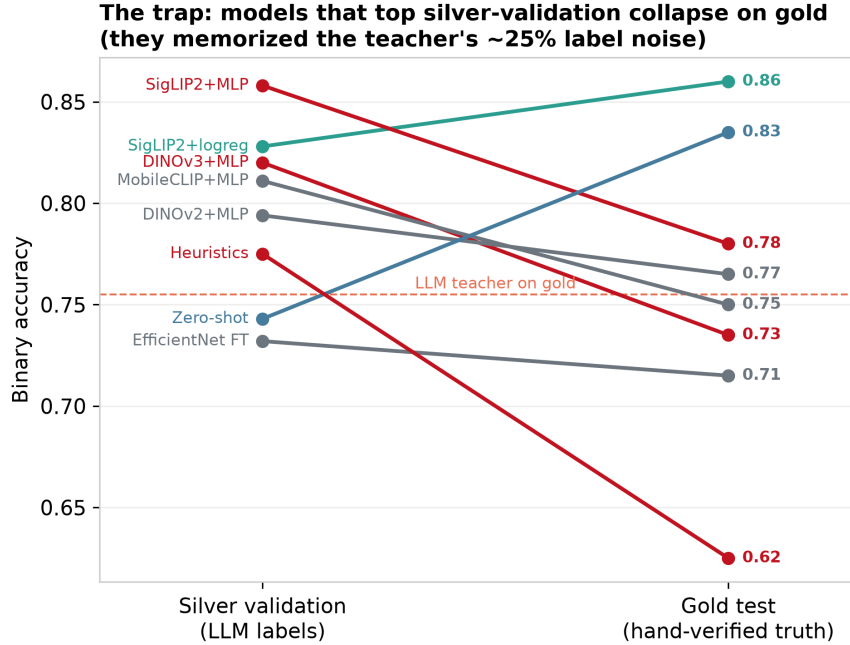


Figure 3: The trap. Ranking on silver validation (LLM labels) selects the model that generalizes *worst* to gold truth.

Page gate — a negative result. An off-the-shelf DocLayout-YOLO page gate scored 61.7% on gold (precision 0.44) and ran at 0.2 pages/s: it detects *figures* but cannot separate a diagram from a plot or photo. The correct page gate reuses the crop classifier: extract figure regions, classify each, mark the page positive iff any crop is a diagram.

Speed. Table 2 and Figure 4 give measured latency on an Apple-Silicon MacBook. The winning gate runs at ~52 ms/crop on GPU (~125 ms CPU), ~100–300× faster than the LLM per decision, free and offline.

Table 2: Measured single-image latency (load → embed → classify).

Component	MPS p50	CPU p50	Peak RAM	Gold acc
Heuristic filter	5.3 ms	5.3 ms	206 MB	0.625
MobileCLIP-S0 + logreg	20.0 ms	168 ms	962 MB	0.750
EfficientNet-B0	27.8 ms	529 ms	799 MB	0.715
SigLIP2 + logreg	51.8 ms	125 ms	1.0 GB	0.860
DocLayout-YOLO (page)	—	~5000 ms	—	0.617
LLM gate (production)	—	4000–17000 ms	—	0.755

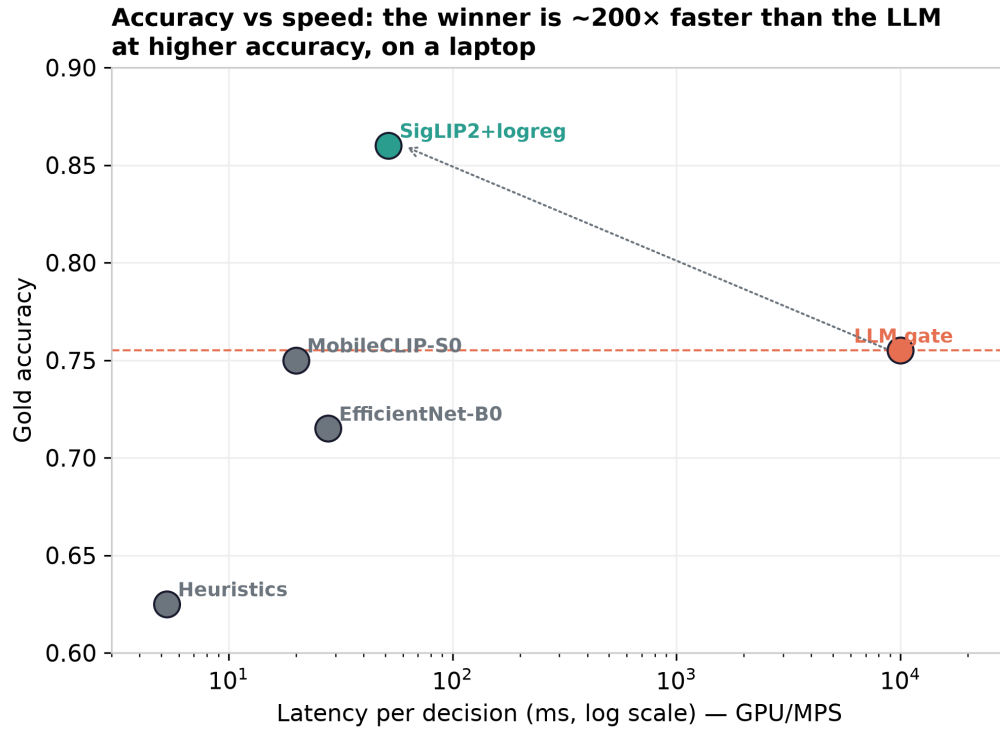


Figure 4: Accuracy vs. latency. The winner sits top-left; the LLM is two orders of magnitude slower at lower accuracy.

Does removing the label noise help? No (Figure 5). Confident-learning removal of the estimated noisiest 4–22% of training labels left the noise-robust winner flat-to-down (86.0%→83–85%) and only let the overfitting MLP catch up, never past the simple baseline. Self-cleaning removes information; it cannot add it. Beating 86% requires independent relabeling or more data, not noise subtraction.

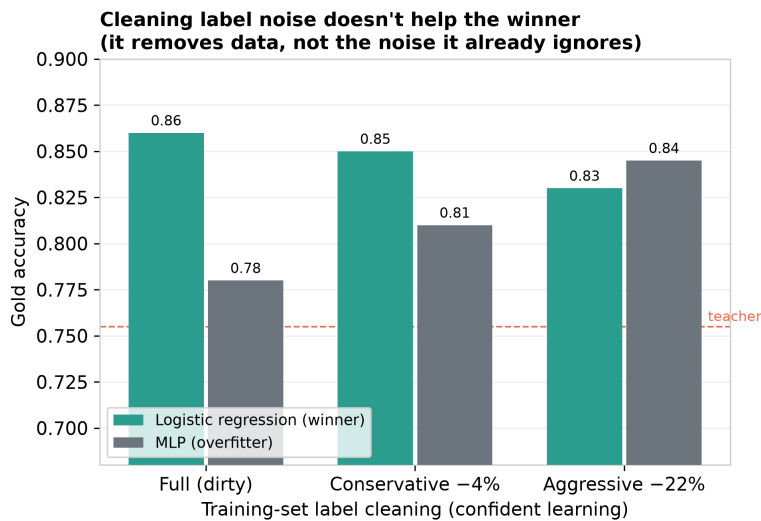


Figure 5: Label cleaning does not help the winner.

5 Production decision

Replace both LLM classification gates with the frozen SigLIP2 + logistic-regression probe. It is more accurate than the LLM (significantly so), $\sim 100\text{--}300\times$ faster, free, offline, and deterministic. Operating threshold 0.55 gives 95.8% accept-precision at full coverage; zero-shot SigLIP2 is the zero-labeled-data fallback, MobileCLIP-S0 (20 ms) the low-latency option. Do not fine-tune and do not use an MLP head at this data scale. The LLM is retained only for *writing* diagram descriptions — and only on crops the free gate already accepted. Rollout is shadow-mode first, then cutover with a 2% LLM audit stream for drift.

6 Limitations

Gold sets are 200 crops / 120 pages, so confidence intervals span a few points and McNemar ties below ~ 5 points. The photo class is small (binary is the reliable metric; 4-class is diagnostic). Robotics is under-represented. Gold annotators and the teacher are all Claude models; the rubric-anchored blind protocol and κ reporting mitigate but do not eliminate shared-family bias.

Artifacts

Code, data manifests, gold labels, per-experiment JSON, and this paper’s sources are released at <https://github.com/adoistic/stem-diagrams>. The 2,000-diagram dataset is available at the project site.

About the author. Adnan Abbasi is the founder of Thothica.